OTH
Amberg-Weiden

AI CONFRENCE

In-context Cascade Segmentation | Hemanthkumar

Amberg, 13. Mai 2026

Übersicht

01 Introduction

02 Background

03 Method

04 Experiments

05 Results

06 Discussion

07 Conclusion

Übersicht

01 | **Introduction**

02 | Background

03 | Method

04 | Experiments

05 | Results

06 | Discussion

07 | Conclusion

Why Medical Image Segmentation Matters

Clinical Importance

- MRI and CT segmentation is essential for:
  - Surgical planning
  - Radiotherapy planning
  - Disease monitoring
  - Treatment evaluation
- Manual annotation is expensive and time-consuming
- Sequential volumes contain many slices requiring expert labeling

Annotation Burden in Medical Imaging

The Core Challenge

Problem

High-quality segmentation requires extensive pixel-level annotations from medical experts.

- Large datasets are difficult to annotate
- Rare diseases have limited labeled data
- Annotation cost limits scalability

Goal: Reduce annotation requirements while maintaining segmentation quality

Sequential Segmentation Problem

Why Slice-wise Inference Fails

- Consecutive MRI/CT slices are strongly correlated
- Standard few-shot methods process each slice independently
- This leads to:
  - Inconsistent boundaries
  - Missing structures
  - Unstable predictions

Übersicht

01 Introduction

02 Background

03 Method

04 Experiments

05 Results

06 Discussion

07 Conclusion

Evolution of Medical Image Segmentation

From CNNs to Foundation Models

- Traditional machine learning methods
- U-Net and encoder-decoder architectures
- Transformer-based segmentation models
- Foundation models for medical imaging

UniverSeg

Few-shot Medical Image Segmentation

- Universal segmentation framework
- Trained on 53 medical imaging datasets
- Performs segmentation without additional retraining

Core Idea

A small support set guides segmentation of unseen query images.

UniverSeg Workflow
Support and Query Relationship

- Support images contain known labels
- Query image is segmented using contextual guidance
- No fine-tuning required

Limitations of Slice-wise Inference

Missing Sequential Context

OTH Amberg-Weiden

Key Limitation

Standard UniverSeg does not propagate information between neighboring slices.

- Each slice is inferred independently
- No memory of previous predictions
- Spatial continuity is not enforced

This motivates the proposed ICS framework.

Übersicht

01 Introduction

02 Background

03 **Method**

04 Experiments

05 Results

06 Discussion

07 Conclusion

In-context Cascade Segmentation (ICS)

Main Idea

Key Concept of ICS

Sequential Information Propagation

1. Start with a small labeled support set
2. Predict neighboring slices sequentially
3. Add predicted masks into the support set
4. Propagate information forward and backward

Main Benefit

Improves inter-slice consistency without retraining.

Forward and Backward Inference

Bidirectional Segmentation

Forward Inference

- Predict subsequent slices
- Update support set continuously

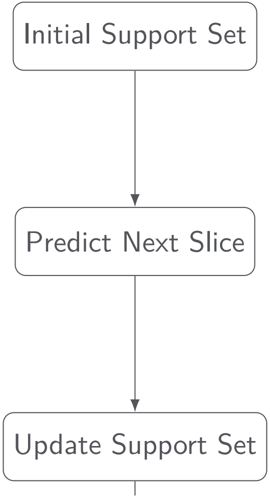
Backward Inference

- Predict previous slices
- Maintain consistency in both directions

ICS Algorithm Workflow

Step-by-step Procedure

OTH
Amberg-Weiden



Why ICS Improves Segmentation

Intuition Behind the Method

- Adjacent slices contain similar anatomical structures
- Previously predicted masks provide contextual guidance
- Sequential propagation stabilizes segmentation boundaries
- Information continuity improves robustness

Übersicht

01 Introduction

02 Background

03 Method

04 Experiments

05 Results

06 Discussion

07 Conclusion

HVSMR Dataset

Cardiac MRI Segmentation Benchmark

- 60 cardiovascular MRI scans
- 8 anatomical regions:
  - LV, RV, LA, RA
  - AO, PA, SVC, IVC

Experimental Setup
Evaluation Protocol

- Baseline: Standard UniverSeg
- Metric: Dice Similarity Coefficient (DSC)
- Support set size varied from 1 to 5
- Forward and backward propagation evaluated

Übersicht

01 | Introduction

02 | Background

03 | Method

04 | Experiments

05 | **Results**

06 | Discussion

07 | Conclusion

Quantitative Results

Dice Similarity Coefficient Comparison

OTH Amberg-Weiden

Observation

ICS significantly improves segmentation performance in several anatomical regions.

Qualitative Results: PA Region

Successful Sequential Consistency

Qualitative Results: LV Region

Failure and Over-segmentation Example

Slice-by-Slice Consistency

Sequential Stability Analysis

Effect of Support Slice Count

Influence of Initial Support Size

Effect of Initial Slice Position

Sensitivity to Initialization

Übersicht

01 | Introduction
02 | Background
03 | Method
04 | Experiments
05 | Results
06 | Discussion
07 | Conclusion

Strengths of ICS

Advantages of the Proposed Method

- Improved inter-slice consistency
- Reduced annotation requirements
- No additional retraining required
- Strong performance in complex anatomical structures

Limitations of ICS

Current Challenges

- Error accumulation during propagation
- Over-segmentation in some regions
- Sensitivity to initial slice selection
- Computational cost with larger support sets

Future Directions

Potential Improvements

- Confidence-aware support updates
- Automated support slice selection
- Validation on CT and ultrasound datasets
- Efficiency optimization and model distillation

Übersicht

01 Introduction

02 Background

03 Method

04 Experiments

05 Results

06 Discussion

07 **Conclusion**

Conclusion
Main Takeaways

1. ICS improves sequential medical image segmentation
2. Propagating pseudo-support information enhances consistency
3. The framework reduces annotation burden while maintaining strong performance

ICS is a promising direction for low-annotation medical segmentation.

Vielen Dank!

Hemanthkumar
Ostbayerische Technische Hochschule Amberg-Weiden
h.muthusamy-gunasekaran@oth-aw.de

Übersicht

08 | Backup Slides

ICS Pipeline Details

Extended Method Overview

Dice Similarity Coefficient (DSC)

Evaluation Metric

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|}$$

- Measures overlap between prediction and ground truth
- $\text{DSC} = 1$ indicates perfect segmentation
- Higher DSC implies better performance

Cardiac Anatomy Labels

HVSMR Dataset Structures

- LV — Left Ventricle
- RV — Right Ventricle
- LA — Left Atrium
- RA — Right Atrium
- AO — Aorta
- PA — Pulmonary Artery
- SVC — Superior Vena Cava
- IVC — Inferior Vena Cava

Support Set Sensitivity

Effect of Support Slice Count

Initialization Sensitivity
Effect of Initial Slice Position

Error Propagation in ICS

Potential Failure Mechanism

OTH Amberg-Weiden

- Incorrect pseudo-labels may propagate errors
- Early segmentation mistakes can accumulate
- Over-segmentation may affect subsequent slices

Computational Considerations

Inference and Efficiency

- ICS avoids retraining and fine-tuning
- Sequential inference introduces additional overhead
- Larger support sets increase computation
- Tradeoff between robustness and efficiency

Comparison with Alternative Approaches
Conceptual Comparison

Method	Sequential	Retraining	Few-shot
U-Net	No	Yes	No
UniverSeg	No	No	Yes
Recurrent Models	Yes	Yes	No
ICS	Yes	No	Yes